INCH-POUND

MIL-DTL-32336B (AR)
17 April 2024

SUPERSEDING
MIL-DTL-32336A (AR)
29 May 2020

DETAIL SPECIFICATION

CAP, BLASTING: NON-ELECTRIC, 30 FOOT SHOCK TUBE - M11, LOAD, ASSEMBLE, AND PACK

This specification is approved for use by the US Army Armaments Center and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification describes the requirements and verification methods for the CAP, BLASTING: NON-ELECTRIC, 30 FT SHOCK TUBE, M11.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-59383	-	Marking Media, Opaque (Porous and Non-Porous Surfaces)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-17124	-	Cord, Detonating and Dummy Cord, Detonating
MIL-DTL-32411	-	Igniter, Time Blasting, Fuse: With Shock Tube Capability - M81, Load, Assemble, and Pack

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	-	Fuzes, Ignition Safety Devices and Other Related Components, Environmental and Performance Tests for
MIL-STD-650	-	Explosive; Sampling, Inspection and Testing
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil .)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US ARMY ARMAMENTS CENTER DRAWINGS

8830953	-	Cup
12983052	-	Packing and Marking for Cap, Blasting: Nonelectric, 30 Foot Shock Tube, M11
13026123	-	J-Hook
13026124	-	Bushing
13026126	-	Shock Tube
13026127	-	Cap, Blasting, Non Electric Assembly
13026128	-	Cap, Blasting: Non-Electric, 30 Ft Shock Tube, M11
13026520	-	Combination of Adopted Items for: Cap, Blasting: Non-Electric 30 Ft Shock Tube, M11 (Total 60 Units/Wood Box)
13045498	-	ESD Shunt

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(Copies of the above drawings are available by email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil.)

US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

ACV00561	-	Unit Load Marking for Shipment and Storage, Ammunition and Explosives
19-48-4116/121R	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Cap, Blasting, Non-Electric, 30 FT. Shock Tube, M11, Packed 60 per Wooden Box, Unitized 9 Boxes per 42” x 53” Pallet; Approx Box size 42-3/4” L x 19” W x 14-1/4” H

(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Default.aspx.)

INTERNATIONAL TEST OPERATING PROCEDURES

ITOP 1-2-601	-	Laboratory Vibration Schedules, International Test Operations Procedure

(Copies of the above international test procedures are available online at https://www.atec.army.mil/publications/topsindex.aspx.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM INTERNATIONAL

ASTM A108	-	Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM D257	-	Standard Test Methods for DC Resistance or Conductance of Insulating Materials

(Copies of the above documents are available online at https://www.astm.org .)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1. Required inspections.

3.1.1 First article. When specified (See 6.2), a sample shall be subjected to the first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2. Materials, components and assemblies. All materials, components and assemblies shall comply with the requirements of Drawing 13026128 and its subsidiary drawings.

3.3. Blasting cap attachment integrity. The blasting cap and shock tube shall not separate and shall function after exposure to an axial force not less than fifteen pounds (15 lbs) for not less than five (5) seconds.

3.4. Tensile integrity. The shock tube shall not exhibit evidence of damage and shall function after exposure to an axial force not less than twenty five pounds (25 lbs) for not less than five (5) seconds.

3.5. Igniter initiation. The M11 shall be capable of initiation upon receipt of a fire signal from an M81 Igniter in accordance with MIL-DTL-32411 and the blasting cap shall function high order (see 6.4).

3.6. Shock tube wall initiation. The M11 shall be capable of initiation through the shock tube wall by using a No. 8 commercial blasting cap and the blasting cap shall function high order (see 6.4).

3.7. J-hook initiation. The M11 shall be capable of initiation utilizing its J-hook and Type I Class E detonating cord in accordance with MIL-DTL-17124 and the blasting cap shall function high order (see 6.4).

3.8. Immersion. The M11 shall function after immersion in water at not less than thirty three and three tenths (33.3) pounds per square inch gage (psig) pressure for not less than two (2) hours.

3.9. Operating temperatures. The M11 shall operate throughout the temperature range of -50ºF to 130ºF.

3.10. Storage temperatures. The M11 shall not exhibit evidence of damage and shall function without performance degradation after being subjected to temperature extremes at -65ºF to 165ºF.

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3.11. Transportation and vibration. The M11 shall not exhibit evidence of damage and shall function without performance degradation after being subjected to transportation vibration at a temperature range of -65˚F to 165˚F.

3.12. Functioning output. After detonation of the blasting cap, the steel block shall have a dent depth of not less than 0.010 inches.

3.13. Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.14. Workmanship. All parts, sub-assemblies, and assemblies shall be clean and free of burrs, chips, sharp edges, cracks, unblended radii, burn marks, checks, sinks, crazes, blisters, excess flash, dirt, grease, oil, rust, corrosion, and other foreign matter.

4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 3 - Examination 2 - Demonstration 4 - Test		Classes of Verification A – First article B – Conformance						
Section 3 and 5 Requirements	Description	Verification Methods				Verification Class		Section 4 Method
		1	2	3	4	A	B	
3.1.1	First article			X	X	X		4.2
3.1.2	Conformance inspection			X	X		X	4.3
3.2	Materials, components and assemblies			X	X	X	X	4.3.4.1 - 4.3.4.7
3.3	Blasting cap attachment integrity				X	X	X	4.4.4
3.4	Tensile integrity				X	X	X	4.4.5
3.5	Igniter initiation				X	X	X	4.4.6
3.6	Shock tube wall initiation				X	X	X	4.4.7
3.7	J-hook initiation				X	X	X	4.4.8
3.8	Immersion				X	X	X	4.4.9
3.9	Operating temperatures				X	X	X	4.4.10
3.10	Storage temperatures				X	X		4.4.11
3.11	Transportation and vibration				X	X		4.4.12
3.12	Functioning output				X	X	X	4.4.13
3.13	Ammunition lot numbering			X		X	X	4.3.3
3.14	Workmanship			X		X	X	4.3.4.1 - 4.3.4.14
5.1	Packaging and marking			X	X	X	X	4.3.4.8 - 4.3.4.14
5.1.1	Barrier bag resistance to leakage				X	X	X	4.5.1, 4.5.2
5.1.2	Unitization			X		X	X	4.3.4.14

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4.1. Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2. First article inspection. When specified, a sample shall be subjected to first article inspection in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures, and facilities as will be used during regular production.

4.2.1 First article rejection. If one (1) or more assemblies, subassemblies, materials, or components fail to comply with the first article requirements, the sample(s) and first article shall be rejected.

TABLE II. First article tests and inspections

Examination or Test	Drawing(s)	Sample Size	Requirement Paragraph	Paragraph Reference / Inspection Method
Cup	8830953-2	30	3.2	4.3.4.1
Cap, blasting, non-electric	13026127-1	30	3.2	4.3.4.2
ESD shunt	13045498	30	3.2	4.3.4.3
Bushing	13026124	30	3.2	4.3.4.4
Shock tube	13026126-3	4.3.4.5	3.2	4.3.4.5
M11 assembly	13026128			
Examination for defects 1/		30	3.2	4.3.4.6
Assembly length		30	3.2	4.3.4.7
Blasting cap attachment integrity		30	3.3	4.4.4
Tensile integrity		30	3.4	4.4.5
Functioning output		30	3.12	4.4.13
Immersion 2/		30	3.8	4.4.9
Storage temperatures 1/	12983052	6 fiberboard boxes	3.10	4.4.11
Operating temperatures		60	3.9	4.4.10
Hot 2/		20	3.9	4.4.10.a
Ambient 2/		20	3.9	4.4.10.b
Cold 2/		20	3.9	4.4.10.c

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TABLE II. First article tests and inspections (continued)

Examination or Test	Drawing(s)	Sample Size	Requirement Paragraph	Paragraph Reference / Inspection Method
Transportation/vibration 1/	12983052	2 wood boxes	3.11	4.4.12
Hot		30	3.11	4.4.12.a
Cold		30	3.11	4.4.12.b
Packaging and marking	13026520	3/	5.1	4.3.4.8 - 4.3.4.14
Fiberboard box (inner)	12983052	3/	5.1	4.3.4.8 – 4.3.4.9
Leakage test		4/	5.1.1	4.5.1
Heat seal seam strength test		4/	5.1.1	4.5.2
Fiberboard box (outer)	12983052	3/	5.1	4.3.4.10 - 4.3.4.11
Wood box	12983052	3/	5.1	4.3.4.12 – 4.3.4.13
Unitized load	19-48-4116/121R	3/	5.1.2	4.3.4.14

Notes:
1/ Indentation represents sequential testing. The sequence shall follow in the order listed.
2/ The immersion and operating temperature samples shall be divided equally as possible into the initiation methods of 4.4.6, 4.4.7, and 4.4.8.
3/ For packaging: The examinations shall be performed on one (1) full pack and one (1) light pack.

For light packs: The light pack shall be no less than 50% filled and the remainder packed out per Drawing 9325874.

For unitized loads: The examinations shall be performed on one (1) sample.

Packaging within unitized load or assembled packaging may include samples from lower level packaging when conformance can still be demonstrated at all levels. With prior Government approval, the samples to be enclosed in the packaging may be substituted with inert mass and volume analogues, reused for each sample, or excluded from packaging if not necessary to demonstrate conformance.
4/ The test shall be performed on three (3) sealed barrier bags from each sealer.

4.3. Conformance inspection.

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4.3.1 Conformance rejection. If any assembly, subassembly, material, or component fails to comply with the conformance inspection requirements of 4.3 and its subparagraphs, the lot from which the sample was taken shall be rejected.

4.3.2 Inspection lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. Inspection lots shall contain materials from not more than one lot interfix number from one manufacturer (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specifications and revisions. In addition each lot shall contain materials from not more than of one lot interfix number from one manufacturer except the following:

a. RDX from not more than two lots from one lot interfix number from one manufacturer.
b. Lead Azide from not more than two lots from one lot interfix number from one manufacturer.

4.3.3 Ammunition lot numbering. Visually verify that a unique ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.3.4 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.14, the definitions of critical, major, and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph “Acceptance by contractor-proposed provisions” of MIL-STD-1916).

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4.3.4.1 Cup.			Drawing Number 8830953-2 Next Higher Assembly 13026127-1	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Cup split or cracked	VL IV	3.2	Visual
102	Outside diameter	VL IV	3.2	Gage
103	Concavity of bottom	VL IV	3.2	Gage
104	Thickness of bottom	VL IV	3.2	Gage
Minor				
201	Inside diameters	VL II	3.2	Gage
202	Workmanship	VL II	3.14	Visual

4.3.4.2 Cap, blasting, non-electric.			Drawing Number 13026127-1 Next Higher Assembly 13026128	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Moisture content of Lead Azide	1/	3.2	4.4.1
102	Total RDX weight	VL IV	3.2	Gage
103	Lead Azide weight	VL IV	3.2	Gage
104	Cup split or cracked	VL IV	3.2	Visual
Minor				
201	Workmanship	VL II	3.14	Visual

Note:
1/ Samples shall be taken from the quantity of Lead Azide used in each shift of Lead Azide loading. For sampling purposes, each daily shift shall be no greater than 10 hours.

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4.3.4.3 ESD shunt.			Drawing Number 13045498 Next Higher Assembly 13026128	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Length	VL IV	3.2	Gage
102	Inner diameter of cavity	VL IV	3.2	Gage
103	Outer diameter at cavity end	VL IV	3.2	Gage
104	Inner diameter	VL IV	3.2	Gage
105	Electrical resistance test	VL IV	3.2	4.4.2
Minor				
201	Workmanship	VL II	3.14	Visual

4.3.4.4 Bushing.			Drawing Number 13026124 Next Higher Assembly 13026128	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Length	VL IV	3.2	Gage
102	Outside diameter	VL IV	3.2	Gage
103	Inside diameter	VL IV	3.2	Gage
Minor				
201	Workmanship	VL II	3.14	Visual

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4.3.4.5 Shock tube.			Drawing Number 13026126-3 Next Higher Assembly 13026128	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Outer diameter	2/	3.2	Gage
102	Tensile strength	1/	3.2	Gage
103	Velocity of detonation	1/	3.2	Gage
104	Shrinkage	1/	3.2	4.4.3
105	Thickness of layers incorrect	2/	3.2	Gage/Visual
Minor				
201	Workmanship	1/	3.14	Visual

Note:
1/ One (1) sample shall be taken from the shock tube process spool at the beginning of each production run in 36in +/- 1/8in length
Accept on zero (0) defects.
2/ Test samples may be taken as needed.

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4.3.4.6 Cap, blasting, non-electric, 30 ft shock tube, M11 - examinations.			Drawing Number 13026128 Next Higher Assembly 12983052	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical				
1	Cup split or cracked, exposed charge, or evidence of explosive leakage	100% + VL VII	3.2	Visual
Major				
101	Length of exposed bushing	VL IV	3.2	Gage
102	Location of 1st crimp from blasting cap edge	VL IV	3.2	Gage
103	Location of 2nd crimp from 1st crimp	VL IV	3.2	Gage
104	Crimp widths (x2)	VL IV	3.2	Gage
105	Crimp diameters (x2)	VL IV	3.2	Gage
106	Shock tube color incorrect	VL IV	3.2	Visual
107	Shock tube is not a continuous strand (contains splices)	VL IV	3.2	Visual
108	Shock tube not free of kinks, twists, and defects	VL IV	3.2	Visual
109	J-hook color incorrect	VL IV	3.2	Visual
110	Shock tube outer diameter (x2)	VL IV	3.2	Gage
Minor				
201	Indicator flag locations	VL II	3.2	Gage
202	Indicator flag colors	VL II	3.2	Visual
203	Label lot number and nomenclature incorrect	VL II	3.2	Visual
204	Workmanship	VL II	3.14	Visual

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4.3.4.7 Cap, blasting, non-electric, 30 ft shock tube, M11 - destructive.			Drawing Number 13026128	
			Next Higher Assembly 12983052	
	Examinations and Tests	Conformance Criteria 1/	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Assembly length	30	3.2	Gage
102	Blasting cap attachment integrity	30	3.3	4.4.4
103	Tensile integrity	30	3.4	4.4.5
104	Functioning output	30	3.12	4.4.13
105	Igniter initiation	2/	3.5	4.4.6
106	Shock tube wall initiation	2/	3.6	4.4.7
107	J-hook initiation	2/	3.7	4.4.8
108	Immersion	30 2/	3.8	4.4.9
109	Operating temperatures	30 2/	3.9	4.4.10
Minor	None defined			

Notes:
1/ Samples from 4.3.4.6 may be used for testing.
The total number of samples is ninety (90).
Accept on zero (0) defects; reject on one (1) or more defects.
The order of sequential testing shall be Majors 101, 102, 103, then 104
2/ The immersion and operating temperature samples shall be divided equally as possible into the initiation methods of 4.4.6, 4.4.7, and 4.4.8.

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4.3.4.8 Packing and marking drawing for the fiberboard box (inner, unsealed).			Drawing Number 12983052 Next Higher Assembly 19-48-4116/121R	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	VL II	5.1	Visual
102	Packaging material moisture content incorrect	1/	5.1	Gage
103	Blasting cap(s) falls out when turned upside down	VL II	5.1	Manual
Minor				
201	Workmanship	VL I	3.14	Visual
Note: 1/ At the start of each production day, three (3) samples of all paper packaging material contained within the barrier bag shall be randomly selected and subjected to moisture content. Any sample(s) and their lot that do not pass the moisture content shall not be used in production for that day.				

4.3.4.9 Packing and marking drawing for the fiberboard box (inner, sealed).			Drawing Number 12983052 Next Higher Assembly 19-48-4116/121R	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Sealing strip torn, badly wrinkled, or otherwise fails to seal box completely or correctly	VL II	5.1	Visual
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1	Visual / Gage
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.14	Visual

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4.3.4.10 Packing and marking drawing for the sealed barrier bags.			Drawing Number 12983052 Next Higher Assembly 19-48-4116/121R	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1	Visual / Gage
102	Leakage test	1/	5.1.1	4.5.1
103	Heat seal seam strength test	1/	5.1.1	4.5.2
Minor				
201	Workmanship	VL I	3.14	Visual
Notes: 1/ The test shall be performed at the beginning of each shift on one (1) sealed barrier bag from each sealer. For sampling purposes, consider a shift to be not more than 10 hours.				

4.3.4.11 Packing and marking drawing for the fiberboard box (outer, sealed).			Drawing Number 12983052 Next Higher Assembly 19-48-4116/121R	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Sealing strip torn, badly wrinkled, or otherwise fails to seal box completely or correctly	VL II	5.1	Visual
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1	Visual / Gage
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.14	Visual

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4.3.4.12 Packing and marking drawing for the wood box (unsealed).			Drawing Number 12983052 Next Higher Assembly 19-48-4116/121R	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	VL II	5.1	Visual
Minor				
201	Workmanship	VL I	3.14	Visual

4.3.4.13 Packing and marking drawing for the wood box (sealed).			Drawing Number 12983052 Next Higher Assembly 19-48-4116/121R	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Box, hinge, or hasp damaged	VL II	5.1	Visual
102	Metallic seal(s) missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	VL II	5.1	4.5.3
103	Markings - including 2D bar codes along with wood preservative and heat treatment markings - missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1	Visual / Gage
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.14	Visual

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4.3.4.14 Unitization.			Drawing Number 19-48-4116/121R Next Higher Assembly	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	VL II	5.1.2	Visual
102	Pallet load markings - including 2D bar codes - missing, incorrect, illegible, or not in proper location	VL II	5.1.2	Visual
103	Filler assembly not built, treated, or marked correctly	VL II	5.1.2	Visual
104	Boxes incorrectly oriented or positioned on pallet	VL II	5.1.2	Visual
105	Strapping not applied correctly or incorrect type	VL II	5.1.2	Visual
106	Unit load not tightly strapped or any strap loose	VL II	5.1.2	Visual/ Manual
Minor				
201	Workmanship	VL I	3.14	Visual

4.4. Test methods.

4.4.1 Moisture content of Lead Azide. The moisture content of the Lead Azide at the time of loading shall be determined in accordance with Method 101.5 of MIL-STD-650, except that a 2.0 gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene. To ensure that the Lead Azide complies with the requirement, adequate controls shall be in place. For verification, the contractor shall select and test a sample from that quantity of Lead Azide used in each shift of loading production up to a maximum of a daily 10 hour shift. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, the Lead Azide represented by the sample shall be rejected. If components have been loaded with Lead Azide containing excessive moisture, the remaining unloaded Lead Azide and the loaded components shall be rejected.

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4.4.2 Electrical resistance test. Measure the resistance of the ESD shunt between Surface A and Surface B in accordance with the ASTM D257. If the electrical resistance is less than 100 or greater than 250,000 OHMs, the sample(s) and lot shall be rejected.

4.4.3 Shrinkage. Expose the sample(s) to 165ºF +/- 5ºF for not less than four (4) hours. At not greater than thirty (30) minutes after conditioning, measure the length of the sample(s) and calculate the shrinkage as a percent of length. The sample(s) and process spool shall be rejected if any sample does not meet the shrinkage requirement.

4.4.4 Blasting cap attachment integrity. Secure the blasting cap. Apply the specified axial load to the junction of the blasting cap and shock tube for the specified length of time. Verify that the blasting cap remains attached to the shock tube . The sample(s) and lot shall be rejected if there is separation of the blasting cap. Upon successful completion, subject the sample(s) to 4.4.5.

4.4.5 Tensile integrity. Secure a portion of shock tube of the M11. Apply the specified axial force to the shock tube for the specified length of time. The sample(s) and lot shall be rejected if any of the shock tube separates. Upon successful completion, subject the sample(s) to 4.4.13.

4.4.6 Igniter initiation. Cut the M11's shock tube within approximately one (1) inch of the environmental seal. Insert the M11 sample following the "Shock tube insertion procedures" and function the M81 following the "Functioning procedures" in accordance with MIL-DTL-32411. Verify the complete detonation of the sample(s).

4.4.7 Shock tube wall initiation. Secure, using commercial tape, a No. 8 commercial blasting cap to the M11's shock tube not greater than four (4) inches from the sealed end. The blasting cap shall make full contact with the shock tube, lengthwise. Initiate the No. 8 commercial blasting cap and verify the complete detonation of the sample(s).

4.4.8 J-hook initiation. Attach detonating cord to the J-hook of the M11 at not greater than twelve (12) inches from the environmental cup. Initiate the detonating cord and verify the complete detonation of the sample(s).

4.4.9 Immersion. Immerse the sample(s) in water at a depth pressure specified not less than two (2) hours. After the removal of the immersed samples, each sample shall be initiated. Verify the complete detonation of the sample(s).

4.4.10 Operating temperatures. Expose the sample(s) of M11 to the following temperatures for not less than four (4) hours:

a.	Hot	125ºF +5/-5ºF
b.	Ambient	75ºF +5/-5ºF
c.	Cold	-45ºF +5/-5ºF

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At not greater than fifteen (15) minutes after removal of the conditioned sample(s) from the chamber, the sample(s) shall be inspected for compliance with the workmanship requirement of 3.14 and initiated. The sample(s) and lot shall be rejected if the M11 does not remain in compliance with the workmanship or functioning requirement.

4.4.11 Storage temperatures. The test shall be conducted on the M11's in their outer fiberboard box. Temperature condition for not less than seventy two (72) hours at the following temperatures:

a. Hot 160ºF +5/-5ºF
b. Cold -60ºF +5/-5ºF

After temperature conditioning, the sample(s) shall be unpackaged and inspected for compliance with the workmanship requirement of 3.14. The sample(s) and lot shall be rejected if the M11 does not remain in compliance with the workmanship requirement.

4.4.12 Transportation/vibration. Perform the transportation/vibration test in accordance with ITOP 1-2-601 on the M11 samples in their wood box(es). Follow ITOP 1-2-601's test procedure of 4.1.1 (Testing of Ammunition). Condition the wood box(es) for not less than twenty four (24) hours to the following temperatures:

a. Hot 160ºF +5˚/-5ºF
b. Cold -60ºF +5˚/-5ºF.

After conditioning, maintain the temperature and conduct the vibration according to the "Wheeled Vehicles" and the "Two-wheeled Trailers" schedules. After not greater than fifteen (15) minutes after the removal of the conditioned samples, each sample shall be inspected for compliance with 3.14 and initiated. Verify the complete initiation of the M11. The sample(s) and lot shall be rejected if the M11 does not remain in compliance with the workmanship or functioning requirement.

4.4.13 Functioning output. Perform output testing per MIL-STD-331 Test D4 (Dent Block Method) on the M11. The dent block shall be made of grade 1018 steel per ASTM A108, have a Rockwell Hardness between B70-B95, and be 1.250 x 1.250 x 0.625 +/- 0.1 inches in size. Secure each blasting cap perpendicularly to a dent block using a test fixture, which only contacts the donor's blasting cap where there is no RDX or Lead Azide and also does not create any confinement on the blasting cap. Initiate the sample and verify the complete detonation of the sample's blasting cap. Verify the dent block measurement. The sample(s) and lot shall be rejected if any blasting cap does not function completely or if any dent block measurement does not meet the requirement.

4.5. Packaging inspection.

4.5.1 Leakage test. The test shall be performed on the sealed barrier bag and its contents (actual or simulated) in accordance with MIL-STD-2073-1 Leakage Test paragraph G.4.2. The test shall follow either the vacuum retention, the submersion (or immersion), or the hot water

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technique. If any sample does not meet the leakage test, any packaged sample(s) in that shift shall be rejected.

4.5.2 Heat seal seam strength test. The test shall be performed on the sealed barrier bag in accordance with MIL-STD-2073-1 Heat seal seam strength test paragraph G.4.3. The samples shall be chosen from the final heat seal seam which closes/seals the pre-assembled bag. If any sample does not meet the heat seal seam test, any packaged sample(s) in that shift shall be rejected.

4.5.3 Metallic seal inspection. The metallic seal shall be inspected for presence, proper markings, and correctness of application. Pull on the metallic seal to determine if it is secure. If the seal breaks the sample and item lot shall be rejected. If the wire breaks before the seal, the box shall be resealed using a new metallic seal.

5. PACKAGING

5.1. Packaging and marking. The materials, components, and assemblies shall comply with all the requirements of Drawing 13026520 and all its subsidiary drawings.

5.1.1 Barrier bag resistance to leakage. The barrier bag shall not leak in accordance with MIL-STD-2073-1 Para G.4.2 (leakage test) and MIL-STD-2073-1 Para G.4.3 (heat- sealed seam test).

5.1.2 Unitization. Loaded pallets shall conform to 19-48-4116/121R. Unit loads shall be marked for logistic traceability in accordance with drawing ACV00561 (Cage 28620).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M11 is part of the Modernized Demolition Initiators (MDI) family. The M11 consists of a military grade blasting cap crimped to a hermetically sealed 30 foot line of shock tube. The M11 is used to initiate all past and current fielded and developmental demolition items such as explosives, shaped charges, and cratering charges. The M11's covered by this specification, are intended for use by military forces to not only initiate demolition items in general but in particular to initiate a multiple number of items. These M11s, covered by this specification are military unique as they must function under rough handling and extreme climatic and storage conditions as encountered by the United States Armed Forces.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2 and applicable documents from Section 6.
b. Requirement/CDRL for Physical Configuration Audit (PCA).
c. Requirement/CDRL for First article inspection.
d. Requirement/CDRL for Conformance inspection.

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e. Requirement/CDRL for Ammunition Data Cards (ADC) as specified in MIL-STD-1168.
f. Requirement/CDRL for Propellants and Explosives data sheet.
g. Acceptance Inspection Equipment (AIE) requirements and associated test procedures. M81's, Detonating Cord, and No. 8 blasting caps (or equivalent) used during FAT and LAT should be submitted as part of the AIE.
h. Requirement/CDRL for Alternate Acceptance Plans
i. Packaging requirements. Special requirements exist for the packaging of this item. Detailed directions and additional information may be obtained from FCDD-ACE-LP.

6.3 Verification Testing. When specified by contract, a sample of the M11 should be subjected to Design Verification Testing in accordance with Drawing 13026866.

6.4 High order detonation. The complete detonation of the M11 is evidenced by the propagation of the shock tube detonation, which initiates the blasting cap. The blasting cap detonates high order, fully consuming all explosive material.

6.5 Critical characteristic. Inadvertent initiation of the blasting cap can lead to injury and death. The blasting cap cup being split or cracked, explosive charge being exposed, or there being evidence of explosive leakage is a critical characteristic of the M11. The explosives contained within the blasting cap are highly sensitive to static discharge and should be contained within the blasting cap cup. Lead Azide is a highly sensitive to impact and static discharge. The cup should not be split or cracked as this may lead to energetic leakage and these defects can lead to injury and death during handling and usage.

6.6 Banding tape. The banding tape's purpose is to consolidate and provide means to identify the M11. Its degradation does not affect the functionality of the item.

6.7 Subject term (key word) listing.
Demolition
MDI (Modernized Demolition Initiators)

6.8 Overview of changes. The following approved Engineering Change Proposal (ECP) is being incorporated into this document: R23Q2020.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army - AR

Preparing activity:
Army-AR
(Project Number: 1375-2023-004)

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